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A Sentiment Analysis of Sexual Assault Reporting by Male and Female Journalists in Uzbek Media

Introduction

This project explores how sexual assault cases are framed and reported by male and female journalists in Uzbekistan. Whether the gender of the author and the source of the sexual assault cases influence the sentiment scores is the main purpose of this study. Sentiment analysis scores range from -1 to 1, where -1 is completely negative, 0 is neutral and 1 is completely positive. My hypothesis is that female journalists tend to use more negative sentiment in news, while male journalists write more negatively in opinion articles. There is a lack of data on sexual assault incidence rate, even by UN Women, for Uzbekistan. Sexual assault often gets normalized in the Uzbek society, where women are usually seen as the cause of the assault, for example, by wearing an exposing clothing and walking in public at night. This study contributes to understanding how linguistic framing in journalism may influence public perceptions of sexual violence.

Previous research has shown that media coverage can shape public perceptions of sexual violence and its victims. 1) Breen et al. (2017) conducted qualitative research on analyzing how Australian news media represent violence against women, dividing independent variables into print and online news. Their study found that routine journalism, like headlines and quotes, made Lazarus (the offender) pitiful while minimizing the victim's voice. 2) Al-Zaman (2023) examined how Bangladeshi media frame rape incidents on Facebook through more than 10, 000 rape-related news headlines posts between 2013 and 2021. They wanted to identify which topics appear most frequently. It turned out judicial processes and characteristics of the victim were the most common topics. The study suggested that people tend to focus

on victim identity rather than seeking to eliminate systematic issues. 3) Chen (2024) investigated how Chinese journalists perceive female victims of sexual violence and how their perceptions might shape news reporting. Their findings indicated that journalists tend to see certain victims as more "ideal," meaning they have more sympathy if the victim is young or weak. The study shows that Chinese journalists' perceptions are shaped by cultural stereotypes and commercial motivations (whether a certain headline gets more clicks).

These studies helped better understand my project, where journalists might alter their voices according to the sensitivity and their own understanding of the issue.

Methodology

The dataset consisted of 60 texts, with 15 observations per group: male-authored newspapers, male-authored articles, female-authored newspapers, and female-authored articles. It was obtained from news sources that are well-known in Uzbekistan: Nemolchi.uz and Qalampir.uz. Each news and article piece was downloaded as a text file. The texts were analyzed using LLMs (ChatGPT 5.2 and ChatGPT 5.2 Thinking) to get a sentiment analysis score ranging from -1 to 1. Words associated with *erkak* (man) vs. *ayol* (woman), on the other hand, were gathered using AntConc, which is a corpus analysis tool. A linear regression model was conducted with the interaction Source (newspaper and article) and GenderAuthor (male and female) as independent factors, while Sentiment score as the dependent variable. The data does not fully meet the assumptions of the multilinear model. The residuals deviate from normality ($p = 1.554e-09$) while the variance is not constant across fitted values ($p = 1.554e-09$). However, the autocorrelation seems to work well ($p = 0.13$). Because of the limited scope of the test, the results of the linear regression model are inconsistent.

	Article	Newspaper
Male	15	15

Female	15	15

Results

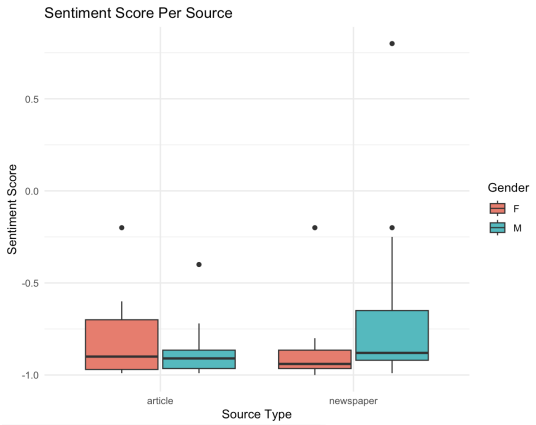

Figure 1

A linear regression model was run to examine whether Source (newspaper vs. article), GenderAuthor (male vs. female), and their interaction predict Sentiment scores in Uzbek media reports on sexual assault. There was no significant main effect of Source (F(1,56)=0.78, p=0.38) and no significant main effect of GenderAuthor (F(1,56)=0.86, p=0.36). The interaction between Source and GenderAuthor almost reached significance (F(1,56)=3.32, β = 0.27, SE = 0.15, t = 1.82, p=0.074). The model explained a small proportion of the variance in sentiment: R^2=0.081, adjusted R^2=0.032. The interaction reveals that the effect of gender authors may depend on the source type. The means were consistent with the hypothesis, but the interaction effect did not reach statistical significance. More broadly, female journalists tend to appear more negative in newspapers, while male journalists are more negative in articles.

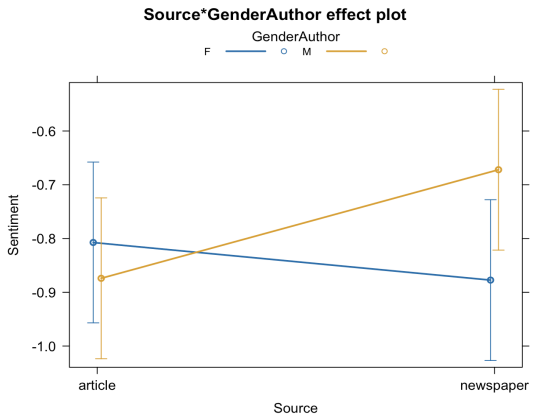

Figure 2

Discussion

The results indicate that gender of authors may shape how sexual assault is represented and framed in Uzbekistan; however, this effect seems to depend on source type. Female journalists tended to use more negative sentiment in newspaper reporting, while male journalists tended to use more negative sentiment in article reporting. The results support my hypothesis, but the evidence is lacking due to limited data. Female journalists might believe that newspapers are read a lot more frequently than articles and, therefore, they try to be assertive in delivering news so the reader feels more emotionally sympathetic toward the issue.

An example from a female author on newspaper reporting: “Proving that a sexual assault has happened is not an easy task. In most situations, the law requires from the victim some strong evidence, but how could a 16-year-old provide that?”

An example from a male author on newspaper reporting: “The Uzbekistan government is trying to implement a lifelong prison for pedophiles. Due to increased violence cases, the Uzbek president Shavkat Mirziyoyev has set a goal to reduce sexual assault on children by 2030.”

Throughout the study, female journalists were framing sentences as moral questions to the reader in hopes to seek better punishment for the perpetrators from the policy makers. I assume that I'm getting differences in the number of sentiments because female journalists write in a varying tone, with words like a criminal, an offender, a pedophile in news as opposed to "X has been accused" by male journalists.

Conclusion

In conclusion, the results show a possible interaction between source type and gender of author in the sentiment analysis of sexual assault cases in Uzbek media. Although the mean values were consistent, the assumptions of the multilinear model were violated, and I predict, it was due to the lack of data and limited scope of the project. The results spotlight how male journalists lean toward their own gender in news reporting. Future work could collect a larger dataset, use other LLMs like Microsoft Copilot, conduct in Russian (since it is widely spoken in Uzbekistan), and use social media posts as a possible source.

Works Cited

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